

Claude Reflect

AI Output Transparency Layer — Architecture & Technology Overview

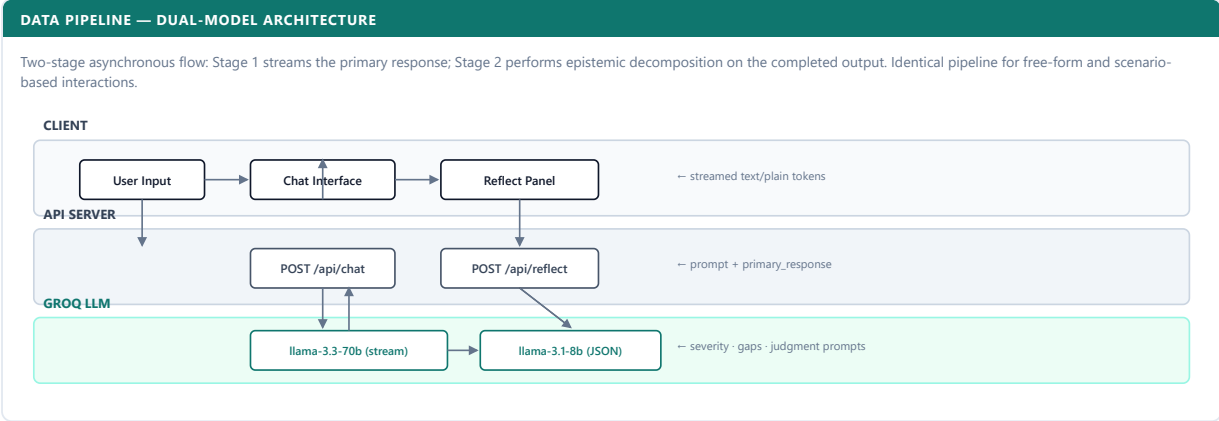
TECHNICAL SPECIFICATION · 1 PAGE

Production claude-reflect-gamma.vercel.app
Repository github.com/parthkapoor2402/claude-reflect
Version 1.0 · June 2026

Overview. Claude Reflect is a full-stack web application that augments generative AI chat with a post-response epistemic analysis layer. A dual-model pipeline separates primary answer generation from structured Reflect output — surfacing assumptions, confidence zones, completeness gaps, and judgment prompts without replacing human decision-making.

CLIENT LAYER	SERVER & AI LAYER
Framework	React 19 (functional components, hooks)
Build tool	Vite 8
Styling	Tailwind CSS 3, CSS variables (theme)
UI / Motion	Framer Motion, Lucide React icons
Rendering	react-markdown (AI responses)
Networking	Fetch API (streaming), Axios

INFRASTRUCTURE	QUALITY ASSURANCE
Hosting	Test framework
Frontend	Jest + Supertest
Backend	Test suite
Serverless Express — <code>api/index.js</code>	18 tests (unit, API, integration)
Routing	Validation
<code>vercel.json</code> (SPA fallback + <code>/api</code>)	<code>reflectValidation.js</code> (JSON normalize)
Local dev	Resilience
concurrently, Vite proxy → :3001	Fallback JSON, 8s client timeout
	Linting
	ESLint (client)



STAGE 1	STAGE 2
Primary Response Generation Endpoint <code>POST /api/chat</code> · Accepts prompt and conversation history · Returns chunked <code>text/plain</code> stream · Powered by llama-3.3-70b-versatile with automatic fallback to 8b model on rate limit.	Epistemic Reflect Analysis Endpoint <code>POST /api/reflect</code> · Accepts prompt and primary response · Returns structured JSON: severity, reasoning foundations, confidence topology, completeness gaps, judgment prompts.